

Customer Segmentation Analysis of a Brazilian E-commerce Dataset

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Abstract

This report details a customer segmentation analysis performed on the Olist Brazilian e-commerce dataset. By applying unsupervised machine learning techniques, specifically K-means and Gaussian Mixture Models (GMM), we aimed to identify distinct customer personas based on their purchasing behavior. The analysis involved significant data pre-processing and feature engineering to create an aggregated, customer-level dataset. Key behavioral metrics similar to an RFM (Recency, Frequency, Monetary) analysis were engineered, including total spending, order count, average review score, and recency. Validation metrics such as the Silhouette Score and Davies-Bouldin Score were used to evaluate clustering performance. The analysis successfully identified four distinct customer segments, for which tailored business actions are recommended to enhance customer engagement, satisfaction, and lifetime value.

1 Data Pre-processing and EDA

The initial step involved loading and merging four key datasets: orders, customers, payments, and reviews. These were joined to create a single comprehensive DataFrame with an initial shape of (103,677, 22), providing a transaction-level view of customer interactions.

1.1 Feature Engineering

To facilitate clustering, the transaction-level data was aggregated to a unique customer level. This process resulted in a new DataFrame with 95,379 rows, where each row represents a unique customer. The following key features were engineered:

- **Total Spending:** The sum of all payment values for each customer.
- **Order Count:** The number of unique orders placed by each customer.
- **Average Review Score:** The mean review score given by a customer.
- **Recency:** The number of days between a customer's last purchase and the latest purchase date in the entire dataset.

1.2 Handling Outliers

An exploratory data analysis of the `total_spending` feature revealed a right-skewed distribution with significant outliers. Figure 1 illustrates this initial distribution. To mitigate the impact of extreme values on the clustering algorithm, outliers were handled by capping them at an upper bound defined by 1.5 times the Interquartile Range (IQR) above the third quartile (Q3).

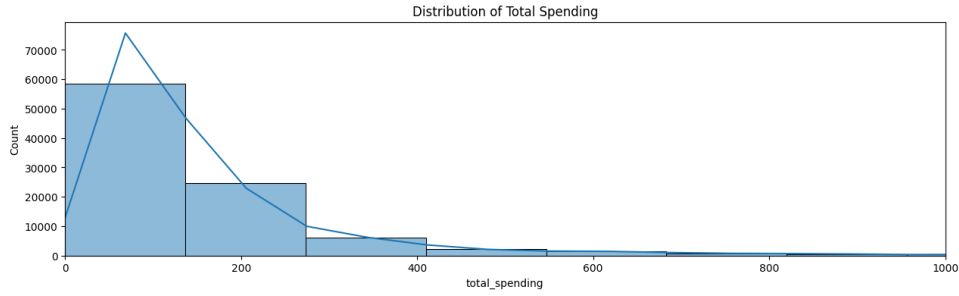


Figure 1: Distribution of Total Customer Spending (Outliers Present).

2 Clustering Model Validation

To group customers into meaningful segments, we implemented and compared two clustering algorithms. All features were scaled using `StandardScaler` before model training.

2.1 Baseline Model: K-means

The Elbow Method was employed to determine the optimal number of clusters (k) for the K-means algorithm. The analysis of the Within-Cluster Sum of Squares (WCSS) indicated that $k=4$ is an appropriate choice, as it represents the "elbow" point where the rate of decrease in WCSS sharply declines (Figure 2).

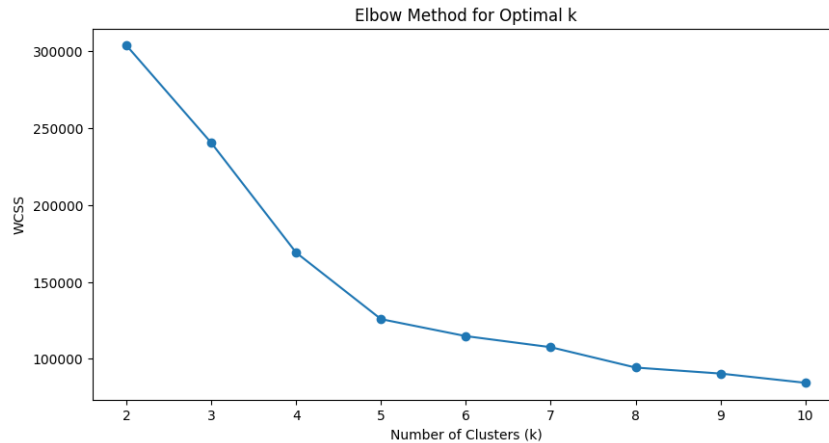


Figure 2: Elbow Method for Optimal k Selection.

2.2 Model Variant: Gaussian Mixture Model (GMM)

As an alternative to K-means, a Gaussian Mixture Model (GMM) was also trained with $k=4$. GMM offers a probabilistic approach to clustering and can capture more complex, elliptical cluster shapes compared to the spherical assumption of K-means.

2.3 Validation Results

The performance of both models was evaluated using the Silhouette Score (higher is better) and the Davies-Bouldin Score (lower is better). While K-means showed slightly superior performance based on these metrics, GMM was chosen for the final cluster profiling due to its flexibility. The results are summarized in Table 1.

Table 1: Clustering Model Validation Metrics.

Model	Silhouette Score	Davies-Bouldin Score
K-means (k=4)	0.3313	1.0061
GMM (k=4)	0.3174	1.0318

Figure 3 visualizes the resulting clusters from both models in a 2D space derived from Principal Component Analysis (PCA).

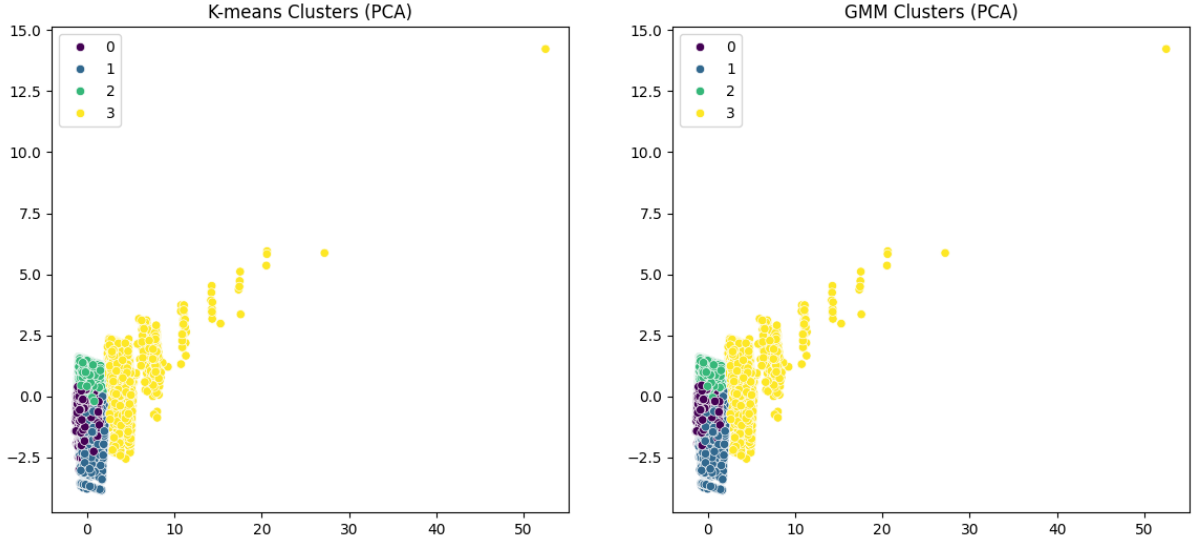


Figure 3: PCA Visualization of K-means (Left) and GMM (Right) Clusters.

3 Cluster Profiles and Recommended Actions

Based on the GMM results, we profiled the four customer segments by analyzing the mean values of their behavioral features (Table 2). This led to the identification of distinct customer personas and actionable business strategies.

Table 2: GMM Cluster Profiles.

Cluster	Total Spending	Order Count	Avg Review Score	Recency	Segment Size
0	\$135.00	1.00	4.74	432.60	31,549
1	\$143.94	1.00	1.86	295.30	21,229
2	\$130.88	1.00	4.76	169.45	39,650
3	\$242.50	2.12	4.12	268.64	2,951

3.1 Cluster 2: Loyal & Recent Customers

- **Portrait:** This is the largest segment, characterized by very high satisfaction (avg. score 4.76) and low recency, indicating they have purchased recently. They are primarily single-purchase customers.
- **Action:** Target with loyalty programs and personalized offers to encourage repeat purchases. Their high satisfaction makes them excellent candidates for providing positive

public reviews.

3.2 Cluster 3: High-Value Repeat Buyers

- **Portrait:** Although the smallest group, these are the most valuable customers, with the highest total spending and an average of over two orders per customer. They are generally satisfied with their purchases.
- **Action:** Treat this segment as VIPs. Offer exclusive deals, early access to new products, and premium customer service to maximize retention and foster brand advocacy.

3.3 Cluster 0: At-Risk / Lapsed Customers

- **Portrait:** This large segment consists of customers who have not purchased in a long time (high recency) but were highly satisfied with their past orders.
- **Action:** Implement re-engagement campaigns. "We miss you" offers, personalized reminders of past purchases, and showcasing new products could effectively bring them back.

3.4 Cluster 1: Dissatisfied Customers

- **Portrait:** A significant segment defined by its extremely low average review score (1.86). Their spending and recency are moderate.
- **Action:** This group requires immediate attention to mitigate brand damage. Conduct targeted surveys to understand the root causes of their dissatisfaction. Offer apologies and potential compensation to attempt service recovery.

4 Limitations and Future Work

- **Limitations:** The analysis was based on a limited set of behavioral features. Important context, such as product category preferences, customer location, and shipping performance, was not included. Furthermore, the model represents a static snapshot and does not account for the evolution of customer behavior over time.
- **Future Work:** The model can be enhanced by incorporating more complex features, such as the average time between orders or sentiment analysis of review text, which could provide deeper insights. A temporal stability test, comparing segments across different time periods (e.g., 2017 vs. 2018), should be conducted to validate the long-term relevance of these personas. Finally, developing an uplift model could help identify which customers within each segment are most likely to respond to specific marketing interventions, thereby optimizing campaign ROI.